# Spring 2007

**Problem 1 (Sp07).** Let $x_1, x_2, \ldots$ be an infinite sequence of real numbers such that every subsequence contains a subsequence converging to 0. Must the original sequence converge?

**Problem 2 (Sp07).** Find a matrix $U$ such that $U^{-1}AU = J$ is in Jordan canonical form, where

$$A = \begin{pmatrix} 0 & -3 & 5 \\ -1 & -6 & 11 \\ 0 & -4 & 7 \end{pmatrix}$$

**Problem 3 (Sp07).** Suppose $f : \mathbb{R} \to \mathbb{R}$ is real analytic and periodic with period $2\pi$. Prove that $f$ has an analytic continuation $F$ defined on a strip

$$S = \{x + iy \in \mathbb{C} \mid |y| < \rho\}$$

with $\rho > 0$, and that $F(z + 2\pi) = F(z)$ for $z \in S$.

**Problem 4 (Sp07).** Define six fields as follows:

- Let $A = \mathbb{Q}(\alpha)$ where $\mathbb{Q}$ is the field of rational numbers and $\alpha$ is the real cubic root of 2.
- Let $B$ be a splitting field of $x^3 - 2$ over $\mathbb{Q}$.
- Let $C$ be an algebraic closure of the field $\mathbb{F}_2$ of 2 elements.
- Let $D$ be the subfield of $C$ generated over $\mathbb{F}_2$ by the set of $a \in C$ such that there exists $n \geqslant 1$ with $a^n = 1$.
- Let $E$ be the field $\mathbb{R}$ of real numbers.
- Let $F$ be the field $\mathbb{Q}[[T]](T^{-1})$ of formal Laurent series with rational coefficients.

For each pair of these, determine whether or not they are isomorphic.

**Problem 5 (Sp07).** Let $a_0(x), a_1(x), \ldots, a_{r-1}(x)$ and $b(x)$ be $\mathcal{C}^m$ functions on $\mathbb{R}$. Prove that if $y'$, $y''$, …, $y^{(r)}$ exists and $y(x)$ is a solution of the differential equation

$$y^{(r)} + a_{r-1}(x)y^{(r-1)} + \cdots + a_1(x)y' + a_0(x)y = b(x)$$

then $y(x)$ is $\mathcal{C}^{m+r}$.

**Problem 6 (Sp07).** Let $A = \alpha_1\sigma_1 + \alpha_2\sigma_2 + \alpha_3\sigma_3$ where $\alpha_1, \alpha_2, \alpha_3 \in \mathbb{C}$ and

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \qquad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \qquad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Let $\beta \in \mathbb{C}$ be any square root of $\alpha_1^2 + \alpha_2^2 + \alpha_3^2$.

1. Prove that $\exp A = \cosh\beta + \dfrac{\sinh\beta}{\beta}A$, where $\dfrac{\sinh\beta}{\beta}$ is interpreted as 1 if $\beta = 0$.

2. Evaluate $\exp A$ explicitly in the case $\alpha_1 = i\pi, \alpha_2 = i\pi$, and $\alpha_3 = \pi$.

**Problem 7 (Sp07).** Let $a$ and $b$ be complex numbers, and let $f : \mathbb{C} \to \mathbb{C}$ be a nonconstant entire function such that $f(az + b) = f(z)$ for all $z \in \mathbb{C}$. Prove that there is a positive integer $n$ such that $a^n = 1$.

**Problem 8 (Sp07).** Let $n \geqslant 3$. Prove that $A_n$ is generated by $(1\ 2\ 3)$ and $(1\ 2 \cdots n)$ if $n$ is odd, or by $(1\ 2\ 3)$ and $(2 \cdots n)$ if $n$ is even.

**Problem 9 (Sp07).** Suppose $b$ and $L$ are positive constants and $f : [0, b] \to \mathbb{R}$ is continuous and satisfies

$$f(x) \geqslant L \int_0^x f(t)\,\mathrm{d}t \qquad 0 \leqslant x \leqslant b$$

Show that $f(x) \geqslant 0$ for $0 \leqslant x \leqslant b$.

**Problem 10 (Sp07).** If $c \in \mathbb{R}$, say that a real valued function $f : \mathbb{R} \to \mathbb{R}$ is periodic with period $c$ if it satisfies $f(x + c) = f(x)$ for all $x \in \mathbb{R}$.

1. Let $V$ be the set of continuous real valued functions $f$ having a positive integer as a period. Prove that $V$ is a vector space.

2. Let $p_1 < p_2 < \cdots < p_n < \cdots$ be the sequence of prime numbers, and for each $i$, let $f_i$ be a function whose minimal positive period is $p_i$. Prove that the functions $f_1, f_2, \ldots$ are linearly independent in $V$.

**Problem 11 (Sp07).** Given any real number $a_0$, define $a_1, a_2, \ldots$ by the rule $a_{n+1} = \cos a_n$ for all $n \geqslant 0$. Prove that the sequence $(a_n)$ converges, and that the limit is the unique solution of the equation $\cos x = x$.

**Problem 12 (Sp07).** Let $k, l$ be positive integers, and $\mathbb{Q}[x]\left(\sqrt[k]{1-x^l}\right)$ any extension field of $\mathbb{Q}[x]$ generated by a k-th root of $1-x^l$. Define $\mathbb{Q}[x]\left(\sqrt[l]{1-x^k}\right)$ similarly. Prove that $\mathbb{Q}[x]\left(\sqrt[k]{1-x^l}\right)$ and $\mathbb{Q}[x]\left(\sqrt[l]{1-x^k}\right)$ are isomorphic.

**Problem 13 (Sp07).** Let $E$ be the complex vector space of entire functions. Let $V$ be a nonzero finite dimensional $\mathbb{C}$-subspace of $E$ with the property that $f \in V$ implies $f' \in V$. Prove that $V$ contains a function that is everywhere nonzero.

**Problem 14 (Sp07).** Let $\mathbb{F}_q$ denote the finite field with $q$ elements, where $q$ is a power of a prime. Let $\mathrm{SL}_n(\mathbb{F}_q)$ be the group of $n \times n$ matrices with entries in $\mathbb{F}_q$ and determinant 1, under matrix multiplication. Determine a simple necessary and sufficient condition on $n$ and $q$ for the center of $\mathrm{SL}_n(\mathbb{F}_q)$ to be trivial.

**Problem 15 (Sp07).** Let $U$ be a nonempty open subset of $\mathbb{R}^d$ and let $f : U \to \mathbb{R}^d$ be a continuous vector field defined on $U$. Let $K$ be a compact subset of $U$ and $b > 0$. Suppose $\varphi : [0, b) \to K$ is a continuous function satisfying

$$\varphi(t) = \varphi(0) + \int_0^t f(\varphi(s))\,\mathrm{d}s \qquad 0 \leqslant t < b$$

Prove that $\lim_{t \to b^-} \varphi(t)$ exists.

**Problem 16 (Sp07).** Given any group $G$, define a binary operation $*$ on the set $H = G \times G$ by $(x, y) * (z, w) = (xz, z^{-1}yzw)$.

1. Show that $(H, *)$ is group.
2. In the case that $G$ is the alternating group $A_n$ on $n$ letters with $n \geqslant 5$, prove that $H$ has no subgroup of index 2.

**Problem 17 (Sp07).** Let $A$ be the set of $z \in \mathbb{C}$ such that $|z| \leqslant 1$, $\Im z \geqslant 0$, and $z \notin \{1, -1\}$. Find an explicit continuous function $u\colon A \to \mathbb{R}$ such that

- $u$ is harmonic on the interior of $A$
- $u(z) = 3$ for $z \in A \cap \mathbb{R}$
- $u(z) = 7$ for $z$ in the intersection of $A$ with the unit circle

**Problem 18 (Sp07).** Let $k$ and $n$ be integers with $n \geqslant k \geqslant 0$. Let $A$ and $B$ be $n \times k$ matrices with real coefficients. For each size-$k$ subset $I \subseteq \{1, \ldots, n\}$, let $A_I$ be the $k \times k$ matrix obtained by discarding all rows of $A$ except those whose index belongs to $I$. Define $B_I$ similarly. Prove that

$$\det(A^\top B) = \sum_I \det A_I \det B_I$$

where the sum is over all size-$k$ subsets $I \subseteq \{1, \ldots, n\}$.